Supervised Learning: Regression Models and Performance Metrics

Question 1: What is Simple Linear Regression (SLR)? Explain its purpose.

Answer:

Simple Linear Regression (SLR) is a statistical method used to model the relationship between one independent variable (X) and one dependent variable (Y).

$$Y = mX + c$$

Where:

- Y = Dependent variable
- X = Independent variable
- m = Slope
- c = Intercept

Purpose of SLR:

Purpose	**Explanation**
Prediction	Predict Y values using X
Relationship Analysis	Understand relationship between X and Y
Trend Identification	Identify trends in data
Forecasting	Estimate future values

Example: Predicting salary based on years of experience.

Question 2: What are the key assumptions of Simple Linear Regression?

Answer:

Assumption	**Meaning**	**How to Check**
Linearity	Relationship is linear	Scatter plot
Independence	Observations are independent	Durbin-Watson test
Homoscedasticity	Constant variance of residuals	Residual plot
Normality	Residuals are normally distributed	Q-Q plot

Question 3: Write the mathematical equation for a simple linear regression model and explain each term.

Equation:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

Explanation of Terms:

Term	**Name**	**Explanation**
Y_i	Dependent Variable	Output variable
X_i	Independent Variable	Predictor variable
β_0	Intercept	Value of Y when X = 0
β_1	Slope	Change in Y for unit change in X
ϵ_i	Error Term	Random error

Question 4: Provide a real-world example where simple linear regression can be applied.

Example: Years of Experience vs Salary

Experience (Years)	**Salary ()**
0	2,50,000
2	3,50,000
4	4,80,000
6	6,00,000
8	7,50,000

Regression equation:

$$Salary = \beta_0 + \beta_1(Experience) + \epsilon$$

Question 5: What is the method of least squares in linear regression?

Answer:

The least squares method finds the best fitting regression line by minimizing the sum of squared residuals.

$$\text{Minimize } \sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

Where:

- Y_i = Actual value
- $\hat{Y}_i$ = Predicted value

Slope formula:

$$\beta_1 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^{n}(X_i - \bar{X})^2}$$

Question 6: What is Logistic Regression? How does it differ from Linear Regression?

Answer:

Logistic Regression predicts probability for binary classification problems.

$$P(Y = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$$

Difference between Linear and Logistic Regression:

Feature	**Linear Regression**	**Logistic Regression**
Output	Continuous values	Probability values
Problem Type	Regression	Classification
Equation	$Y = \beta_0 + \beta_1 X$	Sigmoid equation
Graph Shape	Straight line	S-curve

Question 7: Name and briefly describe three common evaluation metrics for regression models.

Metric	**Formula**	**Description**
MSE	$\frac{1}{n}\sum(Y_i - \hat{Y}_i)^2$	Average squared error
RMSE	$\sqrt{\frac{1}{n}\sum(Y_i - \hat{Y}_i)^2}$	Square root of MSE
MAE	$\frac{1}{n}\sum\lvert Y_i - \hat{Y}_i\rvert$	Average absolute error
R^2	$1 - \frac{SS_{res}}{SS_{tot}}$	Explained variance

Question 8: What is the purpose of the R-squared metric in regression analysis?

Answer:

R-squared measures how much variation in Y is explained by X.

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

Interpretation:

R^2 Value	**Meaning**
0	No explanation
0.5	Moderate explanation
0.8	Strong explanation
1	Perfect fit

Question 9: Python code to fit a Simple Linear Regression model

```
import numpy as np
from sklearn.linear_model import LinearRegression

X = np.array([1,2,3,4,5]).reshape(-1,1)
y = np.array([30,40,50,60,70])

model = LinearRegression()
model.fit(X, y)

print("Slope:", model.coef_[0])
print("Intercept:", model.intercept_)
```

Question 10: How do you interpret the coefficients in a simple linear regression model?

Answer:

Regression equation:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

- β_0 = Intercept
  - Value of Y when X = 0
- β_1 = Slope
  - Change in Y for one-unit increase in X

Example:

$$Salary = 25000 + 5000 \times Experience$$

Interpretation:

- Base salary = 25,000
- Each additional year increases salary by 5,000

Quick Reference

Concept	**Formula**
SLR Equation	$Y = \beta_0 + \beta_1 X + \epsilon$
Least Squares	$\sum(Y - \hat{Y})^2$
R^2	$1 - \frac{SS_{res}}{SS_{tot}}$
Logistic Regression	$\frac{1}{1+e^{-z}}$